

5.4 UDL SEGMENT SHORTER THAN SPAN

5.4.1 Maximum Shear Force

Suppose a udl segment of intensity ω and length l is to traverse a simply supported beam from A to B as shown below. Consider the cross-section X at distance x from A.

The maximum positive and maximum negative shear forces are obtained by positioning the load **strategically** so as to obtain the maximum positive and maximum negative areas under the I.L for V_X .

$$V_{X(\max)(+ve)} = (1/2)[(1 - x/L) + (1 - x/L - l/L)] \omega l$$

$$= \omega l (2L - 2x - l)/2L$$

V_X attains absolute maximum at A, when $x = 0$

$$\text{Thus } V_{X(\max)(abs)(+ve)} = \omega l (2L - l)/2L = V_A$$

$$V_{X(\max)(-ve)} = (1/2)[(x/L - l/L) + x/L] \omega l$$

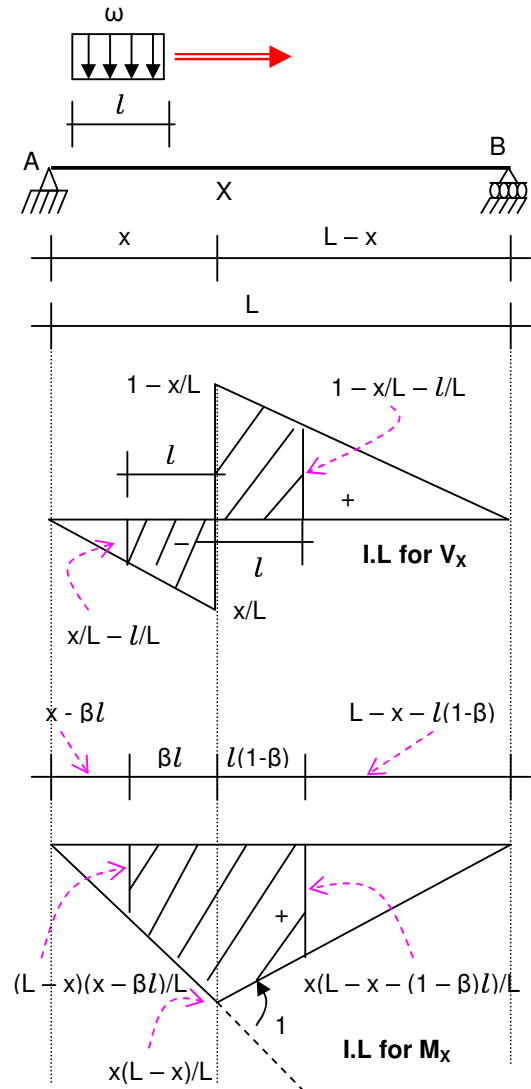
$$= \omega l (2x - l)/2L$$

V_X attains absolute maximum at B, when $x = L$

$$\text{Thus } V_{X(\max)(abs)(-ve)} = \omega l (2L - l)/2L = V_B$$

5.4.2 Maximum Bending Moment

In order to obtain maximum moment the load must be positioned in such a manner that the area under the I. L is maximized.



Let β be the proportion of load to the left of the cross-section X, where $0 \leq \beta \leq 1$.

The shaded area is given

$$A = (1/2)\{(L - x)(x - \beta l)/L + x(L - x)/L\}\beta l + (1/2)\{x(L - x)/L + x(L - x - (1 - \beta)l)/L\}(1 - \beta)l$$

$$A = (1/2)\{2lx - 2lx^2/L - l^2x/L + 2l^2\beta x/L - l^2\beta^2\}$$

$$\text{In order to obtain maximum area, set } \partial A / \partial \beta = 0 \Rightarrow 2l^2x/L - 2l^2\beta = 0 \Rightarrow \beta = x/L$$

Conclusion 1: The maximum bending moment at any point occurs when that point divides the udl segment and the beam span in the same ratio.

$$\beta = \frac{\text{Load to the left of X}}{\text{Total load}} = \frac{\text{Span to the left of X}}{\text{Total span}} = \frac{x}{L}$$

The **absolute maximum bending moment** is obtained by setting $\beta = x/L$ in the expression for A , then $\partial A / \partial x = 0$. This gives $x = L/2$.

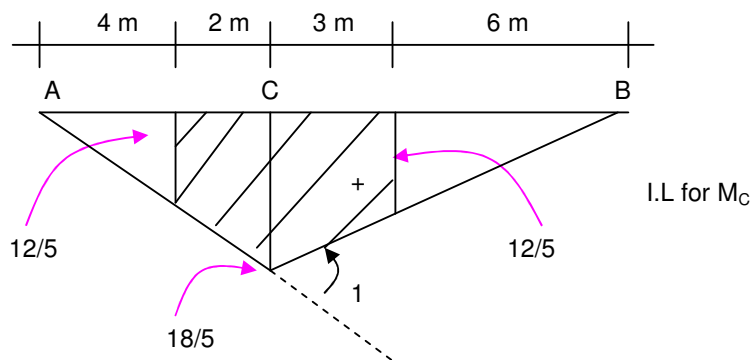
Conclusion 2: The absolute maximum moment occurs at midspan when the udl is centrally positioned there.

Example 5: Complete Example 4.

Solution

$$\omega = 24 \text{ kN/m}, l = 5 \text{ m}, L = 15 \text{ m}$$

$$\text{At C, } x = 6 \text{ m} \Rightarrow \beta = 6/15 \Rightarrow \text{Load to the left of C} = 5 \text{ m} \times 6/15 = 2 \text{ m.}$$



$$\begin{aligned} \text{Shaded Area} &= (1/2) \{ (12/5 + 18/5)x_2 + (12/5 + 18/5)x_3 \} \\ &= (1/2) \{ (12/5 + 18/5)(2 + 3) \} \\ &= 15 \text{ m}^2 \end{aligned}$$

$$M_C (\text{max}) = 24 \times 15 = 360 \text{ kNm}$$